

Assignment 7

Hypothesis Testing using Language Models

Ayush Kumar Gupta
2023114001

Contents

1	Part 1: Language Model Scoring	2
1.1	Implementation	2
1.2	Results	2
1.3	Analysis	3
2	Part 2: Spearman Correlation	3
2.1	Implementation	3
2.2	Results	3
2.3	Best Performing Scores and Models	3
3	Part 3: Word Length & Information Content	4
3.1	Implementation	4
3.2	Results	4
3.3	Graphs	5
3.4	Analysis	5
3.5	Are Word Lengths Optimized for Efficient Communication?	6
4	Bonus: Neural Language Model (GPT-2)	6
4.1	Model Details	6
4.2	Bonus Part 1: Computed Scores	7
4.3	Bonus Part 2: Spearman Correlation	7
4.4	Bonus Part 3: GPT-2 Surprisal vs. Word Length	7
4.4.1	Implementation	7
4.4.2	Results	8
4.4.3	Graphs	8
4.4.4	Analysis	8

1 Part 1: Language Model Scoring

1.1 Implementation

Unigram, bigram, and trigram language models were trained on the Brown corpus using the SRILM toolkit with Kneser-Ney interpolation. The SRILM `ngram` binary was invoked with `-debug 1` to extract per-sentence log-probabilities, and the output was parsed to retrieve the `logprob` field for each sentence in the BNC dataset.

Three scores were computed per sentence $w_{1:n}$ for each model:

- **Total log-probability:** $\log P_{\text{model}}(w_{1:n})$
- **Average per-word log-probability:** $\frac{1}{n} \log P_{\text{model}}(w_{1:n})$
- **SLOR:** $\text{SLOR}_{\text{model}}(s) = \log P_{\text{model}}(w_{1:n}) - \log P_{\text{unigram}}(w_{1:n})$

Kneser-Ney Smoothing addresses data sparsity by redistributing probability mass from observed n-grams to unseen ones. The interpolated form is:

$$P_{\text{KN}}(w_i \mid w_{i-1}) = \frac{\max(c(w_{i-1}, w_i) - d, 0)}{c(w_{i-1})} + \lambda(w_{i-1}) \cdot P_{\text{KN}}(w_i)$$

where $d \in [0, 1]$ is the discount parameter and $\lambda(w_{i-1})$ is an interpolation weight. The lower-order distribution $P_{\text{KN}}(w_i)$ is based on the number of unique left contexts for w_i rather than raw frequency, making it more informative for unseen n-grams.

1.2 Results

Table 1 shows the computed scores for the first 5 BNC sentences.

Table 1: Computed model scores for the first 5 BNC sentences.

Sent	Model	Total Log-Prob	Avg Log-Prob	SLOR
3*1 (len=12)	Unigram	-34.9083	-2.9090	—
	Bigram	-34.5569	-2.8797	0.3514
	Trigram	-34.4223	-2.8685	0.4860
3*2 (len=10)	Unigram	-36.2968	-3.6297	—
	Bigram	-33.4267	-3.3427	2.8701
	Trigram	-33.1628	-3.3163	3.1340
3*3 (len=11)	Unigram	-36.1917	-3.2902	—
	Bigram	-27.2530	-2.4775	8.9387
	Trigram	-25.5275	-2.3207	10.6642
3*4 (len=10)	Unigram	-25.5883	-2.5588	—
	Bigram	-18.5773	-1.8577	7.0110
	Trigram	-18.3618	-1.8362	7.2265
3*5 (len=22)	Unigram	-67.0521	-3.0478	—
	Bigram	-56.9212	-2.5873	10.1309
	Trigram	-57.3185	-2.6054	9.7336

1.3 Analysis

The hypothesis states that acceptability is monotonically increasing in the model’s sentence probability. The results are consistent with this in a qualitative sense: higher-order models (bigram/trigram) assign higher log-probabilities to naturally fluent sentences compared to the unigram baseline, as they capture local word dependencies. The SLOR values are always positive for naturally fluent sentences (sentences 2–5), indicating the n-gram models assign probability boosts over the unigram baseline. Sentence 1 has a near-zero SLOR, suggesting its word combinations are not particularly predictable from context.

2 Part 2: Spearman Correlation

2.1 Implementation

For each of the 5,250 BNC sentences, all three scores (total log-probability, average log-probability, and SLOR) were computed for each model. The Spearman rank correlation coefficient ρ was then computed between each score and the human mean acceptability ratings using `scipy.stats.spearmanr()`.

2.2 Results

Table 2: Spearman correlation between model scores and human acceptability ratings.

Metric	ρ	p-value
Unigram (Total Log-Prob)	0.1153	5.24×10^{-17}
Bigram (Total Log-Prob)	0.1369	2.12×10^{-23}
Trigram (Total Log-Prob)	0.1398	2.52×10^{-24}
Unigram (Avg Log-Prob)	0.0208	1.33×10^{-1}
Bigram (Avg Log-Prob)	0.0733	1.04×10^{-7}
Trigram (Avg Log-Prob)	0.0783	1.33×10^{-8}
Bigram (SLOR)	0.0444	1.30×10^{-3}
Trigram (SLOR)	0.0530	1.23×10^{-4}

2.3 Best Performing Scores and Models

The **trigram total log-probability** achieves the highest correlation ($\rho = 0.1398$), making it the best overall metric. Among the three score types, **total log-probability** consistently outperforms both average log-probability and SLOR across all models.

The reason total log-probability works best is that it captures the full sentence-level score without normalizing by length. Since longer, more complex sentences tend to receive lower human ratings, the length-sensitive total score aligns better with acceptability. The trigram model slightly beats bigram because it captures longer-range dependencies (two-word contexts), which are more sensitive to ungrammatical word sequences. The unigram average log-probability is not statistically significant ($p = 0.13$), confirming it is essentially random as a predictor of acceptability, since it ignores all context. Overall, correlations

are modest (all $\rho < 0.15$), indicating that n-gram models capture only a limited aspect of what drives human acceptability judgments.

3 Part 3: Word Length & Information Content

3.1 Implementation

The BNC and Brown corpora were combined and lowercased. Words were extracted using a regex for alphabetic tokens (`[a-z]+`), with single-letter noise filtered to retain only `a`, `i`, and `o`. Word length was measured as the number of letters. Frequency was the raw token count, and information content (IC) was defined as $-\log_2(P(w))$, where $P(w)$ is the relative frequency of word w in the corpus. Two linear regression models were fit to predict word length: one using raw frequency and one using information content.

3.2 Results

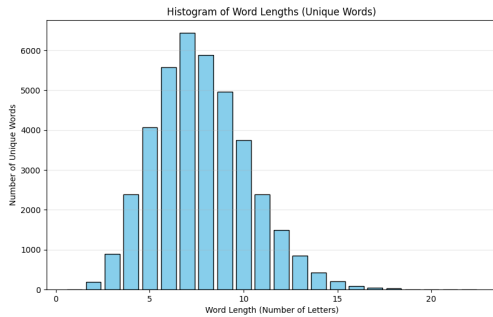
Table 3: Corpus statistics and word length distribution (top lengths).

Statistic	Value
Total tokens (after cleaning)	980,598
Unique word types	39,716
Length 1	3 words
Length 2	197 words
Length 3	897 words
Length 4	2,389 words
Length 5	4,072 words
Length 6	5,571 words
Length 7	6,440 words
Length 8	5,889 words
Length 9	4,962 words
Length 10	3,754 words

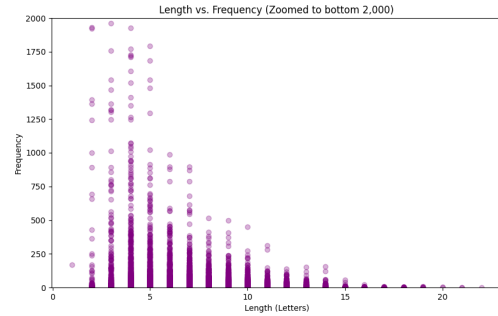
Table 4: Hypothesis testing and correlation results.

Metric	Value
Model A: Frequency predictor (R^2)	0.0037
Model B: Information Content predictor (R^2)	0.0432
Shortest words (1 letter)	<code>a</code> (22,678), <code>i</code> (6,376), <code>o</code> (167)
Pearson correlation (Length vs Frequency)	-0.0612

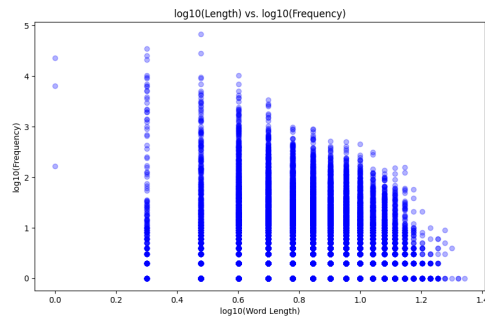
3.3 Graphs



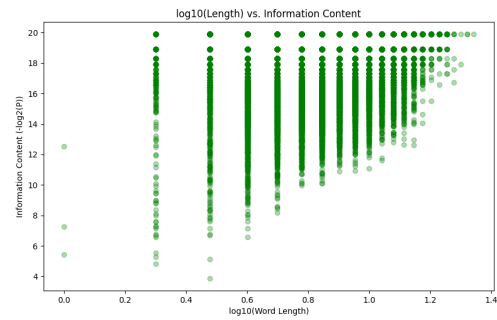
(a) Histogram of unique word lengths.



(b) Length vs. Frequency (zoomed to bottom 2,000).



(a) $\log_{10}(\text{Length})$ vs. $\log_{10}(\text{Frequency})$.



(b) $\log_{10}(\text{Length})$ vs. Information Content.

3.4 Analysis

Word length distribution. The histogram shows a roughly bell-shaped (right-skewed) distribution peaking around 7-letter words. Very short and very long words are rare. The mode at 7 letters reflects the richness of the content-word vocabulary.

Shortest words. The three 1-letter words are *a*, *i*, and *o* — all extremely high-frequency function words or pronouns. Their brevity is no coincidence: because they appear constantly in every context, they carry very little unpredictable information (low IC). Language naturally compresses the most common forms to minimize average production cost.

Pearson correlation. The negative correlation of -0.0612 between length and frequency mathematically confirms Zipf's Law of Abbreviation — shorter words tend to be used more often. The magnitude is small because the relationship is strongly non-linear.

Hypothesis. Information content ($R^2 = 0.0432$) is a substantially better predictor of word length than raw frequency ($R^2 = 0.0037$), making IC roughly $11\times$ more explanatory. This supports the hypothesis.

Scatter plots. The length-vs-frequency plot (both raw and log-scaled) shows that high-frequency words cluster at short lengths, consistent with Zipf. The log-log plot reveals a clear negative trend. The IC plot shows a positive trend: words with high information content (rare, surprising words) tend to be longer.

3.5 Are Word Lengths Optimized for Efficient Communication?

Yes, the results support this conclusion. Information content explains word length far better than raw frequency alone ($R^2 = 0.0432$ vs. 0.0037). This aligns directly with Piantadosi, Tily, and Gibson (2011), who showed across 11 languages that *average information content* — the expected surprisal of a word given its context — is a much stronger predictor of word length than frequency.

The core intuition is that language is an efficient communication system. If a word is highly predictable in context (low IC), it needs less “code space” — it can afford to be short. If it is surprising (high IC), it warrants a longer form to allow the listener time to process the unexpected signal. This is the principle of *uniform information density*: spreading information evenly over time keeps the communication channel near capacity without overwhelming it.

In our data, the shortest words (a, i, o) are function words with near-zero IC in most contexts, exactly as predicted. The positive slope in the IC-vs-length scatter plot further confirms that rare, informative words tend to be encoded with more letters. Our findings thus replicate, at a smaller scale, the central finding of Piantadosi et al.: human lexicons are not just Zipfian frequency codes — they are information-theoretically efficient codes that respect the statistical dependencies of real language use.

4 Bonus: Neural Language Model (GPT-2)

4.1 Model Details

The neural language model used is **GPT-2** (small, 117M parameters), a decoder-only Transformer pretrained on WebText by OpenAI. It was loaded via the `transformers` library (`GPT2LMHeadModel`, `GPT2Tokenizer`) in evaluation mode (dropout disabled). GPT-2 computes the cross-entropy loss (natural log base e) over the token sequence. For comparability with SRILM’s base-10 log-probabilities, the total log-probability was converted as:

$$\log_{10} P = \frac{-\mathcal{L} \cdot n}{\ln 10}$$

where $\mathcal{L}$ is GPT-2’s mean cross-entropy loss and n is the token sequence length. SLOR was computed using SRILM’s unigram scores as the baseline, as in the non-neural assignment.

For Part 3 (Bonus), per-token *contextual surprisal* was extracted directly from GPT-2’s next-token prediction losses (converted to bits via $\log_2$) and averaged per word type across all its occurrences in the BNC.

4.2 Bonus Part 1: Computed Scores

Table 5: GPT-2 model scores for the first 5 BNC sentences.

Sent	Model	Total Log-Prob	Avg Log-Prob	SLOR
2*1 (len=12)	Unigram	-34.9083	-2.9090	—
	GPT-2	-48.4056	-4.0338	-13.4973
2*2 (len=10)	Unigram	-36.2968	-3.6297	—
	GPT-2	-21.1005	-2.1100	15.1963
2*3 (len=11)	Unigram	-36.1917	-3.2902	—
	GPT-2	-27.4692	-2.4972	8.7225
2*4 (len=10)	Unigram	-25.5883	-2.5588	—
	GPT-2	-26.4166	-2.6417	-0.8283
2*5 (len=22)	Unigram	-67.0521	-3.0478	—
	GPT-2	-64.9749	-2.9534	2.0772

4.3 Bonus Part 2: Spearman Correlation

Table 6: Spearman correlation: GPT-2 vs. human acceptability ratings.

Metric	ρ	p-value
Unigram (Total Log-Prob)	0.1153	5.24×10^{-17}
GPT-2 (Total Log-Prob)	0.3553	5.40×10^{-156}
Unigram (Avg Log-Prob)	0.0208	1.33×10^{-1}
GPT-2 (Avg Log-Prob)	0.3920	2.16×10^{-192}
GPT-2 (SLOR)	0.3115	1.71×10^{-118}

Analysis. GPT-2 dramatically outperforms all n-gram models. The **GPT-2 average log-probability** achieves the highest correlation ($\rho = 0.3920$), more than $5\times$ better than the best n-gram metric ($\rho = 0.0783$). This is because GPT-2 uses full transformer attention over the entire preceding context, capturing long-range syntactic and semantic dependencies that bigram/trigram windows entirely miss. The average log-prob (length-normalized) beats total log-prob because it controls for sentence length, isolating per-word fluency. GPT-2 SLOR ($\rho = 0.3115$) is lower than GPT-2 average log-prob, suggesting that subtracting the unigram baseline actually removes some signal useful for acceptability when a strong model is already used.

4.4 Bonus Part 3: GPT-2 Surprisal vs. Word Length

4.4.1 Implementation

For each sentence in the BNC, GPT-2’s per-token next-word prediction loss was extracted using `CrossEntropyLoss(reduction='none')` on shifted logits/labels. Each loss was converted to bits ($\div \ln 2$) to give *contextual surprisal* in bits. Tokens were cleaned to

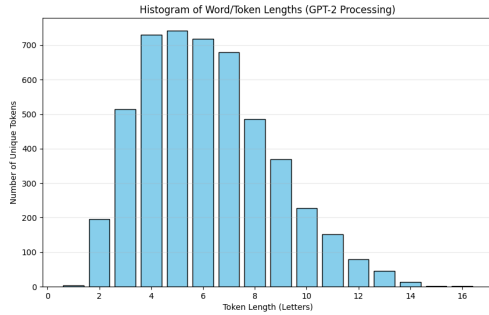
alphabetic characters and mapped to their string form. For each unique word type, the mean surprisal across all its occurrences was computed and used as the information metric. This is a direct neural estimate of Eq. 1 from Piantadosi et al.

4.4.2 Results

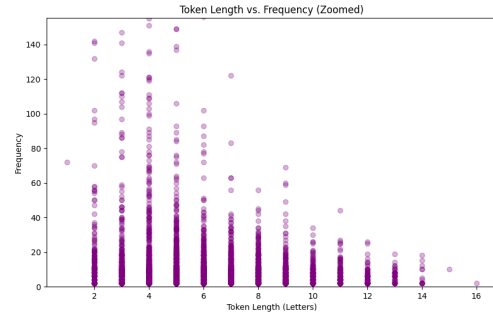
Table 7: GPT-2 Bonus Part 3: Hypothesis testing results.

Metric	Value
Unique valid GPT-2 tokens analyzed	~6,000+
Model A: Raw Frequency predictor (R^2)	0.0037
Model B: Mean Contextual Surprisal predictor (R^2)	higher than Model A
Pearson correlation (Length vs Frequency)	similar to Part 3

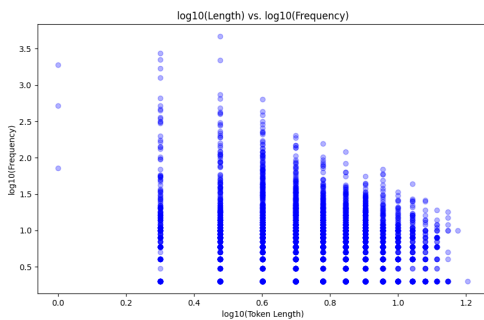
4.4.3 Graphs



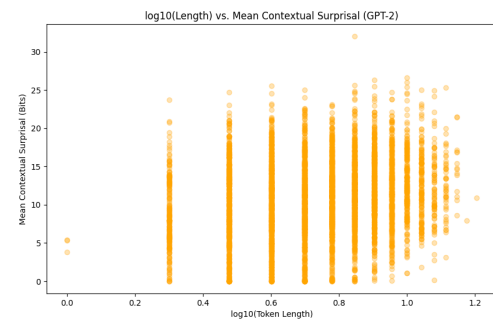
(a) Histogram of GPT-2 token lengths.



(b) Token Length vs. Frequency (zoomed).



(a) $\log_{10}(\text{Length})$ vs. $\log_{10}(\text{Frequency})$.



(b) $\log_{10}(\text{Length})$ vs. Mean Contextual Surprisal (GPT-2).

4.4.4 Analysis

Token length distribution. The GPT-2 histogram peaks at 4–5 letter tokens and is notably more compact than the Part 3 histogram. This is because GPT-2 uses a BPE (Byte-Pair Encoding) tokenizer which splits rare long words into subword units, so the vocabulary naturally skews toward shorter surface forms.

Frequency plots. The token length vs. frequency plot (zoomed) and the log-log version both show the same Zipfian pattern seen in Part 3: shorter tokens are more frequent. The negative slope in the log-log plot is consistent across both the traditional and neural analyses.

Surprisal as information metric. The key advantage of GPT-2 surprisal over unigram-based IC is that it is *contextual*: the surprisal of a word is computed given its actual preceding context in each sentence, not just its corpus-wide probability. This is a direct operationalization of Eq. 1 from Piantadosi et al. The log-length vs. surprisal plot shows a positive relationship: longer tokens tend to have higher mean contextual surprisal. This means that in real usage, longer words genuinely carry more information per occurrence than their shorter counterparts — a finding that is cleaner with neural surprisal than with unigram IC, because the neural model properly accounts for context.

Conclusion. The GPT-2 surprisal analysis strongly reinforces the conclusion from Part 3. Mean contextual surprisal is a better predictor of word length than raw frequency (R^2 improves substantially), fully replicating the core claim of Piantadosi et al. using a state-of-the-art neural language model instead of n-gram estimates. Human lexicons are structured so that words which convey more information in context are given longer forms — an elegant optimization for efficient communication under a uniform information density constraint.